

20 Mortality and morbidity

10 primary questions | Oldest to newest

2026 objectives: 4.2.2

8 related questions are filed elsewhere in this volume. Find them by selecting this chapter and enabling related topics in [index.html](#).

Click a question below to begin.

CA1-P1 September 2009 Q4	14 marks
CA1-P1 April 2016 Q5	13 marks
CA1-P1 September 2016 Q5	16 marks
CA1-P1 September 2017 Q6	18 marks
CA1-P1 April 2018 Q6	27 marks
CP1-P1 April 2019 Q8	30 marks
CP1-P1 April 2020 Q1	2 marks
CP1-P1 April 2020 Q7	12 marks
CP1-P1 April 2023 Q10	21 marks
CP1-P1 September 2023 Q7	17 marks

CA1-P1 | September 2009 | Q4

Current topic | 14 marks total | Original PDF p2

Related: Ch 18, Ch 23.

- 4** A life insurance company in a certain country has been selling term assurances directly to policyholders via the internet for five years. The company is about to conduct an analysis of the mortality experience on this portfolio.
- (i) Describe the considerations that should be taken into account when deciding how the data should be grouped. [6]

This analysis shows that the mortality experience on this portfolio is much lighter than what was allowed for in the pricing basis.

- (ii) Discuss possible reasons for this. [3]

CA1-P1 | September 2009 | Q4 - continued

[Current topic](#) | [14 marks total](#) | [Original PDF p3](#)

Related: Ch 18, Ch 23.

Currently, the company has different premium rates for males and females. The government of this country is about to introduce legislation banning gender discrimination in the provision of goods and services. There are no exceptions for insurance.

- (iii) Discuss the effect this legislation will have on the approach that this company uses to set premium rates. [5]

[Total 14]

CA1-P1 | April 2016 | Q5

Current topic | 13 marks total | Original PDF p3

Related: Ch 22, Ch 2.

5 An insurance company writes regular premium term assurance policies which are used in conjunction with mortgage loans.

- (i) Outline why the assumptions used for mortality rates will be important in ensuring that the policies are correctly priced. [4]

The government of the country that the insurance company operates in has recently announced that a recreational drug (i.e. a drug used for pleasure not for medical reasons) that is currently illegal will be de-criminalised. That is, the current ban on its possession, consumption or supply will be completely lifted.

This decision has been taken after extensive independent research. This research has concluded that the harmful health effects of the drug on its users are not as serious as previously thought, although there is still some uncertainty about the longer term effects of using the drug.

- (ii) Discuss how the change in policy relating to the drug could affect the mortality experience under the term assurance policies written by the insurance company. [6]

- (iii) Describe the issues that may arise in relation to the Terms and Conditions covering the term assurance policies. [3]

[Total 13]

CA1-P1 | September 2016 | Q5

Current topic | 16 marks total | Original PDF p3

Related: Ch 18, Ch 19.

- 5** A life insurance company sells annuity business. It has been investigating its mortality experience.

The investigation has shown a significantly higher number of deaths over the last three years relative to those expected under the existing mortality assumptions.

The investigation has also concluded that the higher number of deaths has been due to smoking related illnesses.

In light of this experience, the insurance company is considering adjusting the mortality assumptions it uses to price its annuity business.

- (i) Discuss the data limitations that the insurance company will need to consider when deciding whether the results of the investigation are suitable for adjusting the mortality assumptions used to price its annuity business. [8]

Approximately 20% of the business written over the last three years has been for individuals taking out policies with very large annuities.

- (ii) Discuss the data sources that could be used to set mortality assumptions for these individuals. [5]
- (iii) Suggest, with reasons, whether the mortality assumptions for this “large annuity” subsection of the business should differ from those used for “standard” annuity business. [3]
- [Total 16]

CA1-P1 | September 2017 | Q6

Current topic | 18 marks total | Original PDF p4

Examiner report and original paper: use the links for this question in [index.html](#).

- 6** There has been a distinctive pattern to mortality rates by age in a country for many years.

Male mortality rates from ages five to seventeen are low and similar. Thereafter, they rise steeply, peaking at age twenty five. They then fall steadily to age thirty where the mortality rate is similar to that at age nineteen.

Female mortality rates from ages five to thirty are lower than male rates and generally rise steadily with age without the peak at age 25 prevalent in male mortality rates.

- (i) Discuss why the mortality rates between genders described above might vary so significantly. [5]

Recent anecdotal mortality evidence from the last three years appears to imply that female mortality rates are changing and now show a similar pattern to male mortality rates from ages seventeen to thirty.

- (ii) Outline why it may be inappropriate for the providers of death benefits to change the assumptions they use for female mortality rates in the light of this evidence. [5]

Subsequently, detailed investigations have shown that male mortality rates have not changed significantly, but female mortality rates from ages seventeen to thirty are moving closer to male mortality rates.

- (iii) Discuss the possible implications of this change in female mortality rates for:
- (a) the premium charged by a motor insurance company.
 - (b) the premium charged by an insurance company that writes term assurance policies used in conjunction with mortgage loans.
 - (c) the pension contributions required from a retailer of women's clothes which provides pension and death benefits for its employees through a defined benefit pension scheme.

[8]

[Total 18]

CA1-P1 | April 2018 | Q6

Current topic | 27 marks total | Original PDF p4

Related: Ch 18, Ch 23.

- 6** In a developed country, a life insurance company writes only term assurance business. It is the largest seller of term assurance business in the country.

The country's medical profession has recently announced that a small number of people have become the first lives, in that country, to be diagnosed with a particular disease that until recently has been confined purely to a small number of developing countries on another continent.

In those developing countries, the disease has been classed as infectious and has led to a significant increase in the mortality rate in those countries.

- (i) Discuss the factors which the insurance company should consider in relation to whether its new term assurance policies should be re-priced in response to the medical profession's announcement. [6]

Following this development the insurance company has decided to re-price its policies.

- (ii) Discuss the data that could be used to set revised mortality assumptions following the medical profession's announcement. [14]
- (iii) Describe how the remaining pricing assumptions are likely to be affected. [4]
- (iv) Suggest other important areas of the insurance company's business (other than pricing) that the insurance company may now need to keep under closer review. [3]

[Total 27]

CP1-P1 | April 2019 | Q8

Current topic | 30 marks total | Original PDF p4

Related: Ch 38.

- 8**
- (i) Explain why it is necessary to have different mortality tables for different classes of lives. [5]
 - (ii) Discuss, in each of the following separate cases, the expected differences in the level of mortality:
 - (a) term assurance policyholders versus general population
 - (b) an employer's group life policy with an insurance company versus the insurance company's term assurances
 - (c) pensioners of an employer's directors' pension scheme versus pensioners of the employer's staff pension scheme. [18]
 - (iii) Discuss why it is important to monitor mortality experience. Your answer should include references to the cases in part (ii). [7]
- [Total 30]

CP1-P1 | April 2020 | Q1

[Current topic](#) | [2 marks total](#) | [Original PDF p2](#)

Examiner report and original paper: use the links for this question in [index.html](#).

- 1** Explain the concept of mortality convergence. [2]

CP1-P1 | April 2020 | Q7

Current topic | 12 marks total | Original PDF p3

Related: Ch 18.

- 7** A large corporation has purchased a number of companies from different sectors of the economy of the country it operates in. Each company has a defined benefit pension scheme. The past mortality experience has varied significantly between each of the defined benefit pension schemes.

- (i) Outline the factors that could explain the differing mortality experience. [2]
- (ii) Describe the types of information that would be required to investigate the main factors affecting the mortality experience of a scheme. [7]

The members of a scheme are allowed to move their accrued pension benefits out of the scheme. The scheme's actuary calculates a lump sum 'Transfer Value' to be paid out by the scheme for this purpose.

- (iii) Outline the issues of using the types of information in (ii) to calculate Transfer Values for a scheme. [3]

[Total 12]

CP1-P1 | April 2023 | Q10

Current topic | 21 marks total | Original PDF p5

Related: Ch 6.

- 10** A life assurance company sells a wide range of immediate annuities, including impaired life annuities. Lives in poor health are able to apply for an impaired life annuity. One of the rating factors used is the postal code (also known as the zip code) of the annuitant.
- (i) Describe why the company will use homogeneous groups when pricing its annuities. [5]
 - (ii) Outline the reasons why an individual may seek to purchase an immediate annuity. [3]
 - (iii) Suggest, with reasons, the main types of life who are likely to be priced on an individual basis. [2]
 - (iv) Discuss how useful the postal code may be as a means of understanding some of the principal factors that contribute to the variation in mortality of the non-impaired annuitants. [6]
 - (v) Outline the possible types of selection among the non-impaired annuity lives. [5]
- [Total 21]

CP1-P1 | September 2023 | Q7

Current topic | 17 marks total | Original PDF p4

Related: Ch 0, Ch 18.

7 A life insurance company has sold term assurance and whole life assurance business for many years.

- (i) Describe the various types of selection that can be expected to occur in respect of this portfolio of business. [8]

The company is now also considering selling immediate annuity business.

- (ii) Outline how the actuarial control cycle can be used to consider the risks and potential profitability of this new product line. [3]

- (iii) List the possible external sources of information the life insurance company could consult before deciding to sell immediate annuity business. [3]

- (iv) Explain why using the data from the term assurance and whole life assurance business may be inappropriate in setting the longevity assumptions for the immediate annuity business. [3]

[Total 17]